SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1005	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
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Date:	09/10/2026	**Duration:**	60 minutes

Code of Conduct

I ***KARNAM HARSHITH (192511384)*** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _____

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. **Understand the Problem Statement**
   - Explain the problem in your own words.
   - Identify the objectives of the proposed system.
   - State the scope of the problem.
2. **Identify Key Stakeholders and Challenges**
   - Identify all the stakeholders involved in the system.
   - Explain the roles and responsibilities of each stakeholder.
   - Discuss the major challenges and issues faced by the stakeholders.
3. **Analyze the Current Approach**
   - Describe the existing system or current process.
   - Identify its strengths and limitations.
   - Analyze the problems or gaps that need to be addressed.
4. **Propose a Suitable Solution Using Software Engineering Principles**
   - Design a software-based solution for the given problem.
   - Describe the major functional modules of the proposed system.

- Identify the functional and non-functional requirements.
- Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.

5. **Justify the Chosen Methodology/Framework**
   - Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
   - Justify why the selected methodology/framework is suitable for the proposed system.
   - Explain how it supports successful project development and maintenance.
6. **Discuss Expected Outcomes and Limitations**
   - Describe the expected benefits of the proposed solution.
   - Explain how the solution addresses the identified challenges.
   - Discuss potential risks, implementation challenges, and limitations.
   - Suggest possible future enhancements.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering	Proposes an innovative, feasible solution applying appropriate software	Suitable solution with good application of software	Basic solution with limited application of software	Methodology is inappropriate, poorly justified, or not discussed.

Principles (15%)	engineering principles (modularity, scalability, maintainability, security, etc.).	engineering principles.	engineering principles.	
Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	**Mark**
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25

CO1 – AT1

CASE STUDY ANALYSIS

QUESTION 19

INTELLIGENT IT HELP DESK AND SERVICE MANAGEMENT PLATFORM

Case Study on Intelligent IT Service Management

Submitted By

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INTELLIGENT IT HELP DESK AND SERVICE MANAGEMENT PLATFORM

1. PROBLEM UNDERSTANDING

1.1 Problem Statement

Organizations depend on IT infrastructure, software applications, networks, computers and digital services for daily operations. Employees frequently face problems such as network connectivity issues, software failures, hardware problems, password/account issues and application errors.

In many organizations, employees report these problems through different channels such as email, phone calls, messages or direct communication with IT staff. This makes it difficult to systematically record, categorize, prioritize, assign and track service requests. Important requests may be delayed or overlooked, while IT staff spend additional time manually following up on problems. SE CO1 AT 1

To overcome these limitations, an **Intelligent IT Help Desk and Service Management Platform** is proposed. It provides a centralized platform where employees can create service tickets. The system supports ticket categorization, prioritization, team assignment, status tracking, SLA monitoring, knowledge-base support, feedback and analytics.

1.2 Objectives

The main objectives are:

- Provide a centralized platform for reporting IT problems.
- Create and maintain service tickets.
- Categorize problems into network, hardware, software and account issues.
- Prioritize incidents according to urgency, severity and impact.
- Assign tickets to appropriate IT support teams.
- Track ticket status from creation to closure.
- Monitor SLA response and resolution deadlines.
- Maintain a knowledge base for common problems.
- Collect employee feedback.
- Generate reports and analytics.
- Reduce IT service processing and resolution time. SE CO1 AT 1

1.3 Scope

The system manages the complete lifecycle of an IT service request, beginning with problem reporting and ending with resolution, feedback and ticket closure.

Major functions include:

1. User and role management
2. Ticket creation and management
3. Problem categorization
4. Incident prioritization
5. Support team assignment
6. Ticket status tracking
7. SLA monitoring
8. Knowledge base
9. User feedback
10. Analytics and reporting

1.4 Need for the Proposed System

Traditional IT support becomes difficult to manage when the number of requests increases. Common problems include fragmented reporting, manual processing, poor visibility, incorrect prioritization, missed SLA deadlines and repeated troubleshooting. A centralized system can reduce manual effort, improve transparency and provide better service-performance information. SE CO1 AT 1

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2. STAKEHOLDER AND CHALLENGE ANALYSIS

2.1 Stakeholders

Stakeholder	Role	Major Requirements
Employees	Report and track IT problems	Easy ticket creation, tracking, notifications
IT Support Staff	Analyze and resolve problems	Assigned tickets, priority, SLA deadlines
IT Administrator	Manage system	Users, roles, teams and configuration
IT Manager	Monitor IT services	SLA monitoring, reports and performance
Management	Evaluate IT performance	Analytics, trends and service reports

The original case study identifies these five major stakeholder groups and their requirements. SE CO1 AT 1

2.2 Major Stakeholder Challenges

Accurate Categorization

Employees may describe the same problem differently. For example, "Wi-Fi is not working" and "unable to connect to office network" may represent the same network issue.

Correct Prioritization

Critical incidents affecting many employees should receive higher priority than minor individual issues. Priority should consider severity, urgency, business impact and number of affected users.

Appropriate Team Assignment

Problem	Assigned Team
Network Issue	Network Support Team
Hardware Problem	Hardware Support Team
Software Problem	Application Support Team
Account Problem	System Administration Team

Incorrect assignment can increase resolution time. SE CO1 AT 1

SLA Monitoring

Support requests may have response and resolution deadlines. Manual monitoring can result in missed SLA targets. The system should track deadlines and generate alerts.

Security

Different users require different access levels:

- Employee → Own tickets
- Support Staff → Assigned/authorized tickets
- IT Manager → Reports
- Administrator → System managementAuthentication, authorization and role-based access control are therefore important. SE CO1 AT 1

3. CURRENT SYSTEM ANALYSIS

3.1 Existing System

The existing IT support process generally uses email, phone, messaging or direct communication. IT staff manually review requests, determine their category and priority, assign them to appropriate personnel and communicate the resolution.

Existing Workflow

Employee → Reports Problem → Email/Phone/Message → IT Support → Manual Review → Manual Categorization → Manual Priority → Team Assignment → Troubleshooting → Resolution → Closure

3.2 Strengths

- Simple communication.
- Direct interaction between employees and IT staff.
- Low initial setup cost.
- Flexible reporting.
- Suitable for organizations with a small number of requests.

3.3 Weaknesses

- No centralized ticket management.
- Manual categorization.
- Manual prioritization.
- Difficult ticket tracking.
- Poor SLA monitoring.
- Repeated troubleshooting.
- Limited performance analysis.
- Requests may be lost or overlooked.

3.4 Gap Analysis

Existing Problem	Proposed Improvement
Scattered requests	Centralized ticket management
Manual categorization	System-assisted categorization
Manual priority assignment	Structured priority mechanism
Manual team assignment	Systematic team assignment
Manual deadline monitoring	Automated SLA monitoring
No organized knowledge base	Centralized knowledge base

Existing Problem	Proposed Improvement
Limited reporting	Analytics dashboard
Limited status visibility	Real-time ticket tracking

These gaps show the need for centralized ticket management, systematic categorization, priority management, team assignment, SLA monitoring, knowledge management, tracking and analytics. SE CO1 AT 1

4. PROPOSED SOLUTION USING SOFTWARE ENGINEERING PRINCIPLES

4.1 Proposed System

The proposed **Intelligent IT Help Desk and Service Management Platform** centralizes the complete IT service-request lifecycle.

Employees submit problems as tickets. The system supports categorization and prioritization, assigns tickets to suitable support teams and monitors SLA deadlines. IT staff investigate and resolve problems, while managers can monitor performance through reports and analytics. SE CO1 AT 1

Proposed Workflow

Employee → Login → Create Ticket → Enter Problem Details → Categorize → Prioritize → Assign Team → SLA Monitoring → IT Investigation → Status Update → Resolution → Closure → Feedback → Analytics

4.2 Major Components

1. Authentication and Role Management
Provides secure login and role-based access for employees, support staff, managers and administrators.

2. Ticket Management
Employees create tickets containing problem details. Tickets move through:

New → Assigned → In Progress → Resolved → Closed

3. Intelligent Categorization
Problems are categorized as network, hardware, software or account-related issues.

4. Ticket Prioritization
Priority is determined using urgency, severity, impact and affected users.

5. Team Assignment
Tickets are routed to the appropriate support team according to their category.

6. SLA Monitoring
The system tracks response and resolution deadlines and provides alerts for approaching or overdue tickets.

7. Knowledge Base
Common technical solutions are stored so employees and support staff can find solutions to frequently occurring problems.

8. Feedback and Analytics
Employees can provide feedback after resolution, while managers can analyze ticket volume, resolution time, SLA compliance, team performance and user satisfaction.

4.3 Software Engineering Principles

Modularity

The system is divided into modules such as authentication, ticket management, SLA management, knowledge base and analytics. This allows independent development and testing.

Maintainability

A structured architecture allows future changes, such as adding new ticket categories, without redesigning the entire system.

Scalability

The system should support increasing numbers of users and tickets as the organization grows.

Security

Authentication, authorization, role-based access control, input validation, secure database access and password protection should be used.

Usability

The interface should be simple enough for employees without advanced technical knowledge.

Reliability

Database transactions, validation, error handling and backup mechanisms help maintain accurate ticket information and prevent loss of service requests.

Reusability

Common components such as authentication, notifications, validation and reporting can be reused across the platform. SE CO1 AT 1

4.4 Benefits

The proposed system provides:

- Centralized IT service management.
- Faster categorization and assignment.
- Better prioritization of critical incidents.

- Improved ticket visibility.
- Better SLA monitoring.
- Reduced repetitive troubleshooting.
- Improved communication.
- Better service-performance analysis.
- Improved user satisfaction.
- More organized IT support operations. SE CO1 AT 1

5. METHODOLOGY / FRAMEWORK JUSTIFICATION

5.1 Agile Scrum

The proposed system will follow the **Agile Scrum methodology**. Agile Scrum develops software through short development cycles called **Sprints**. Each Sprint develops, tests and improves a specific set of features.

This methodology is suitable because help desk requirements may change according to feedback from employees, IT support staff and managers. SE CO1 AT 1

5.2 Why Agile Scrum?

- Supports changing requirements.
- Allows incremental development.
- Encourages continuous stakeholder feedback.
- Supports continuous testing.
- Identifies problems early.
- Allows important features to be developed first.
- Improves team communication.
- Reduces development risk.

5.3 Sprint Plan

Sprint	Main Activities	Output
Sprint 1	Requirements, roles, authentication	Login and role management
Sprint 2	Ticket creation and status	Ticket management
Sprint 3	Categorization, priority, assignment	Intelligent ticket routing
Sprint 4	SLA and notifications	SLA tracking
Sprint 5	Knowledge base and feedback	Knowledge and feedback
Sprint 6	Analytics and testing	Complete integrated system

This sprint structure is based on the original case-study plan. SE CO1 AT 1

Scrum Development Cycle

Product Backlog → Sprint Planning → Sprint Backlog → Design & Development → Testing → Sprint Review → Stakeholder Feedback → Retrospective → Next Sprint

Scrum Roles

- **Product Owner:** Defines and prioritizes requirements.
- **Scrum Master:** Facilitates Scrum and removes obstacles.

- **Development Team:** Designs, develops, tests and integrates the system.

Agile Scrum is appropriate because the platform contains multiple interconnected modules and requirements may evolve through user feedback. SE CO1 AT 1

6. EXPECTED OUTCOMES AND LIMITATIONS

6.1 Expected Outcomes

The proposed system is expected to improve efficiency, transparency and reliability of IT support operations. SE CO1 AT 1

Faster Ticket Processing

Centralized ticket creation, categorization, prioritization and assignment can reduce processing time.

Better Ticket Tracking

Employees can check ticket status without repeatedly contacting IT support.

Improved SLA Compliance

The system can monitor deadlines and alert support staff about approaching or overdue tickets.

Reduced Repetitive Troubleshooting

The knowledge base provides solutions for frequently occurring problems.

Improved IT Productivity

Reducing manual ticket tracking and assignment allows support staff to focus more on problem resolution.

Better Management Decisions

Analytics can provide information about ticket trends, categories, resolution time, SLA compliance, team performance and user satisfaction. SE CO1 AT 1

6.2 Key Performance Indicators

KPI	Purpose
Average Resolution Time	Measures how quickly tickets are resolved
Average Response Time	Measures support response speed
SLA Compliance Rate	Measures tickets resolved within SLA
Ticket Resolution Rate	Measures successful resolution
User Satisfaction	Measures employee satisfaction
Reopened Ticket Rate	Measures quality of resolution
Ticket Volume by Category	Identifies common IT problems

6.3 Limitations

1. **Incorrect Categorization** – vague descriptions may require human verification.
2. **Data Quality** – incorrect or incomplete ticket information can affect classification and analytics.
3. **Knowledge Base Maintenance** – solutions must be regularly reviewed and updated.
4. **Initial Setup Effort** – roles, categories, priority rules, SLA policies and support teams must be configured.
5. **User Adoption** – employees and IT staff must consistently use the platform.
6. **Security Risks** – sensitive organizational information requires strong security controls.
7. **Integration Challenges** – integration with existing IT systems may require additional development.
8. **Scalability Requirements** – large organizations may require additional infrastructure and database optimization. SE CO1 AT 1

6.4 Future Enhancements

Future versions can include:

- AI-powered conversational chatbot.
- Natural-language ticket classification.
- Predictive identification of recurring incidents.
- Automated resolution recommendations.
- Mobile application.
- Email and communication-system integration.
- Enterprise ITSM integration.
- Advanced predictive analytics.
- Automated escalation and notifications. SE CO1 AT 1

CONCLUSION

The **Intelligent IT Help Desk and Service Management Platform** provides a structured approach to IT service management. It centralizes ticket management, categorizes and prioritizes incidents, assigns requests to appropriate teams, monitors SLAs, maintains a knowledge base and provides analytics.

The system can improve IT support efficiency, transparency and user satisfaction. However, its success depends on data quality, user adoption, knowledge-base maintenance, security and proper configuration. Continuous evaluation and improvement based on user feedback and service performance are therefore necessary. SE CO1 AT 1